GRADE 7 ACCELERATED · UNIT 6 · PRACTICE TEST

Relationships in Triangles

18 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Triangle Inequality and the Converse

3 points MA.8.GR.1.3

1. Which set of side lengths can form a triangle? *1 pt*

- ○ A. 4 cm, 7 cm, 11 cm
- ○ B. 12 cm, 5 cm, 6 cm
- ● **C. 6 cm, 8 cm, 13 cm**
- ○ D. 3 cm, 10 cm, 15 cm

C

The two shorter sides against the longest: 6 + 8 = 14, more than 13. A lies flat; B passes only when the first two listed are added; D's two shorter sides cannot reach across 15.

Scoring: 1 point.

2. Select ALL of the sets of side lengths that make a right triangle. *1 pt*

- ☒ **A. 17, 8, 15**
- ☐ B. 5, 6, 7
- ☒ **C. 20, 21, 29**
- ☐ D. 7, 24, 26
- ☐ E. 4, 6, 8

A and C

Square all three and test the longest as c: 64 + 225 = 289 = 17^2 (the longest listed first) and 400 + 441 = 841 = 29^2. Both and nothing else for the point.

Scoring: 1 point.

3. Two sides of a triangle are 6 inches and 15 inches long. Which inequality shows every possible length x, in inches, of the third side? *1 pt*

- ○ A. $6 < x < 15$
- ● **B.** $9 < x < 21$
- ○ C. $9 \le x \le 21$
- ○ D. $0 < x < 21$

B

15 − 6 = 9 and 15 + 6 = 21, never equal to either. A stays between the two given sides; C counts the flat cases, 9 and 21; D has only the sum.

Scoring: 1 point.

The Pythagorean Theorem

6 points MA.8.GR.1.1

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

4. Find the missing side of each right triangle.

a. Its legs are 9 inches and 12 inches. How long is the hypotenuse? *1 pt*

15 in

$c^2 = 81 + 144 = 225$

b. Its hypotenuse is 61 meters and one leg is 60 meters. How long is the other leg? *1 pt*

11 m

a leg: subtract — 3,721 − 3,600 = 121. Adding the squares gives a 'leg' longer than the hypotenuse.

Scoring: a 1 · b 1 = 2 points.

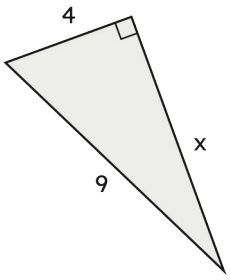
5. A right triangle has legs of 7 centimeters and 13 centimeters. How long is the hypotenuse, in centimeters? Round your answer to the nearest tenth. *1 pt*

14.8 cm (exactly √218 cm)

$c^2 = 49 + 169 = 218$; √218 = 14.76…

Scoring: 1 point.

6. Which equation can be used to find x? *1 pt*



- ○ A. $4 + x = 9$
- ○ B. $4^2 + 9^2 = x^2$
- ○ C. $4^2 + x^2 = 9$
- ● **D.** $4^2 + x^2 = 9^2$

D

The 9 is across from the right angle, so it is c and stands alone: $x^2 = 81 − 16 = 65$. A adds the sides, not their squares; B puts the hypotenuse with the legs; C leaves the 9 unsquared.

Scoring: 1 point.

7. A rectangular garden measures 9 meters by 10 meters. A path runs straight across it from one corner to the opposite corner.

a. How long is the path, in meters? Round your answer to the nearest tenth. *1 pt*

13.5 m (exactly √181 m)

the path is the hypotenuse: 81 + 100 = 181

b. How much shorter is the path than walking along two sides of the garden? Round your answer to the nearest tenth. *1 pt*

5.5 m

9 + 10 = 19 m along two sides; 19 − 13.5. Follow-through on part (a).

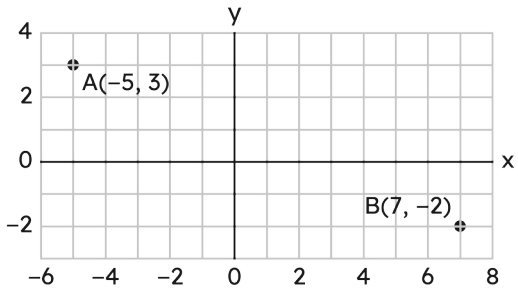
Scoring: a 1 · b 1 = 2 points.

Distance on the Coordinate Plane

6 points MA.8.GR.1.2

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

8. Find the distance between points A and B. *1 pt*



13 units

Across 7 − (−5) = 12, down 3 − (−2) = 5: c^2 = 144 + 25 = 169.

Scoring: 1 point.

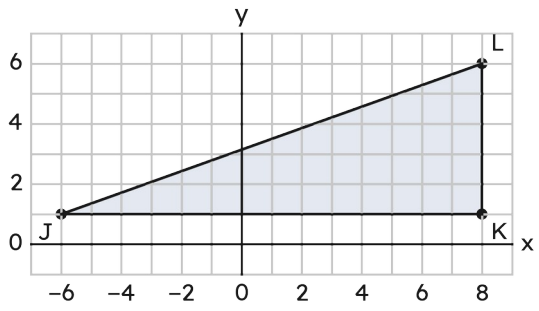
9. Find the distance between (−1, 3) and (4, 1). Round your answer to the nearest tenth. *1 pt*

5.4 units (exactly √29)

Across 5, down 2: c^2 = 25 + 4 = 29.

Scoring: 1 point.

10. Triangle JKL has corners J(−6, 1), K(8, 1) and L(8, 6).



a. Find the length of side JK. *1 pt*

14 units

Across zero: 6 + 8 — never 8 − 6.

b. Find the perimeter of triangle JKL. Round your answer to the nearest tenth. *1 pt*

33.9 units

KL = 5; JL = √(196 + 25) = √221 ≈ 14.866; 14 + 5 + 14.866 = 33.866 — round once, at the end. Follow-through on part (a).

Scoring: a 1 · b 1 = 2 points.

11. On a town map, each unit is one block. The library is at (−1, −5) and the pool is at (8, 7).

a. How far is it from the library to the pool along the streets? *1 pt*

21 blocks

The streets follow the grid: 9 across and 12 up.

b. How far is it in a straight line? *1 pt*

15 blocks

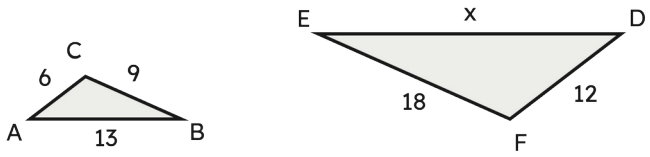
The hypotenuse of legs 9 and 12: 81 + 144 = 225.

Scoring: a 1 · b 1 = 2 points.

Similar Triangles

3 points MA.8.GR.2.4

12. Triangle ABC is similar to triangle DEF. Find x. *1 pt*

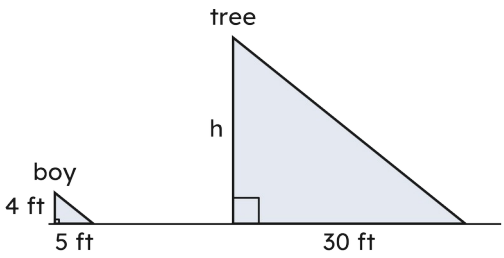


x = 26

The names match AC with DF and BC with EF — 12 with 6 and 18 with 9, as by size — so k = 2, and x is on DE, which matches AB: 13 × 2. 39 matched by place on the page (18 with the 6); 13 took similar as equal; 19 added the difference 12 − 6.

Scoring: 1 point.

13. A boy 4 ft tall casts a shadow 5 ft long. At the same moment, a tree casts a shadow 30 ft long. How tall is the tree? *1 pt*



24 ft

k = 30 ÷ 5 = 6 from the shadows; 4 × 6. 37.5 matched height and shadow crosswise; 120 multiplied by the boy's height, not by k.

Scoring: 1 point.

14. Which triangle is similar to a triangle with sides 6, 7 and 10? *1 pt*

- ● **A. 12, 14, 20**
- ○ B. 10, 11, 14
- ○ C. 12, 14, 18
- ○ D. 6, 7, 12

A

Every side × 2: 12/6 = 14/7 = 20/10. B adds 4 to every side; C passes two ratios but not the third, 18 ÷ 10; D keeps two sides the same.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.8.GR.1.3	1, 2, 3	3	
MA.8.GR.1.1	4, 5, 6, 7	6	
MA.8.GR.1.2	8, 9, 10, 11	6	
MA.8.GR.2.4	12, 13, 14	3	
Total		**18**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.